Understanding Support Vector Machines in Machine Learning: A Study of Kernels and Regularisation

Machine Learning Indiviual Assignment

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Git Hub Link: https://github.com/kotakalyan/Machine-Learning-Individual-assignment.git

Introduction

Support Vector Machines (SVM) are a type of supervised machine learning algorithms used mainly for classification problems. The main goal of the SVM algorithm is to find a hyperplane that best separates the data between classes. While simple classification approaches try to separate the data, SVMs do this by trying to maximise margin between classes, which helps improve generalisation in unseen data.

The foundation of SVM is hyperplane, which serves as the decision boundary. The algorithm chooses this hyperplane such that it is at the farthest distance from the closest data point of all classes. These key exemplars are referred to as support vectors, and they are what allow us to identify not only the placement of the boundary but also its orientation. (And SVMs are especially robust, since the model depends solely on a small subset of training data.)

This paper explores the present behaviour of Support Vector Machines via experiments. We start with the linear SVM to build some intuition and move on to non-linear classification via kernel methods later on using, for instance, Radial Basis Function (RBF) Kernel. The impact of regularisation is also examined and a real world example is discussed to illustrate the effectiveness of SVMs on complicated, non-linear separable datasets

Linear Support Vector Machines

The first and simplest kind of Support Vector Machine is the linear SVM, where you separate your data with a straight-line decision boundary. This hyperplane separates the feature space into two regions corresponding to different classes. The main goal of the model is definitely not to get any separating line, but the one that maximizes the margin between two classes.

Hence, margin is defined as the distance from the data points of each class and the best hyperplane. The points closest to the hyperplane are called support vectors, and they are what determine where the optimal boundary would sit. SVM, unlike many other models, does not rely on all data points but only the ones that lie near the margin which reduces its sensitivity to noise hence more robust and efficient.

Figure 1: Linear SVM showing the decision boundary, margins, and support vectors.

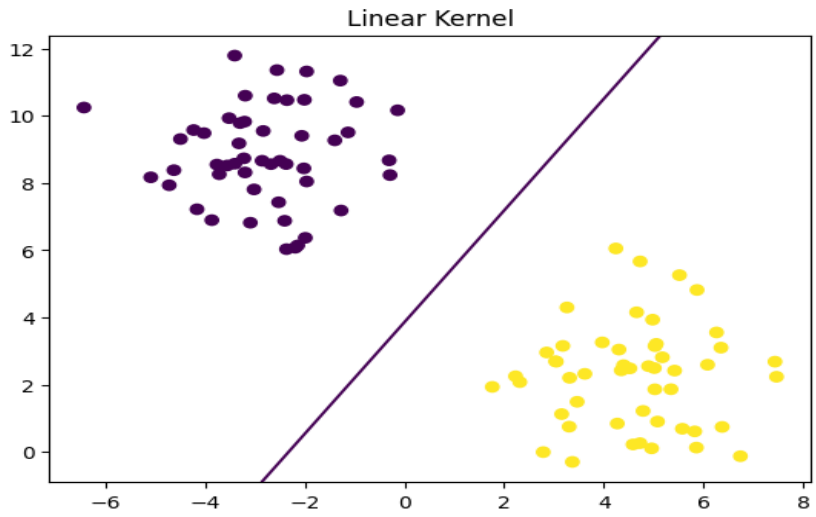
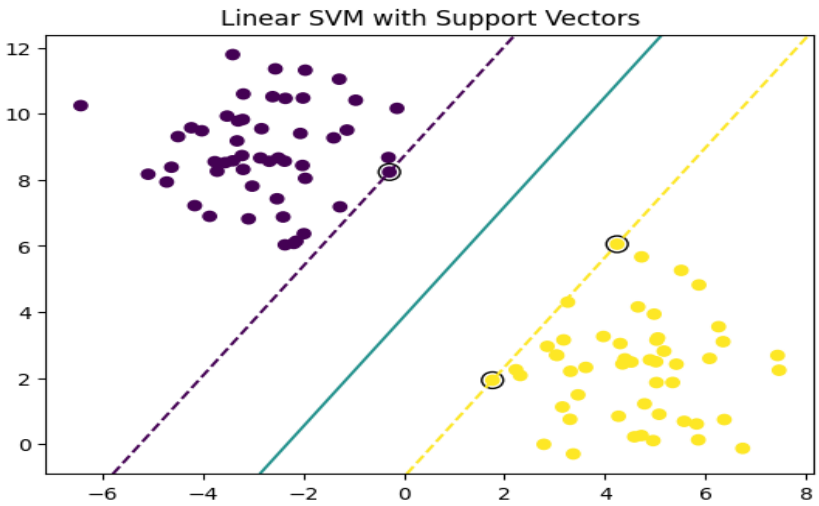


Figure 2: Linear SVM showing decision boundary, margins, and support vectors.



The linear SVM separates the dataset with a straight decision boundary (see Figure 1). The solid line defines on the optimal hyperplane, and the dashed lines represent the margins in either direction. The support vectors are the points that lie closest to the boundry and effect its position — this is illustrated by the circled points

This works fine if the data is linearly separable, i.e., a straight, perfect line separates the classes. For many real-world applications though, our data is not this simple and cannot easily be separated by a linear dividing line. Such a shortcoming motivates the investigation of more sophisticated methods, for example kernel ones covered in the next section.

Non-linear Classification and the Kernel Trick

Although linear SVMs work well for simple datasets that can be separated by a straight line, many real-world scenarios include more complex patterns with non-linear separation. In other words, a basic SVM model struggles to identify the relevant decision boundary and misclassifies the data points in such cases.

Support Vector Machines solve this by using a method known as kernel trick. 356) Instead of trying to directly separate the data in its original feature space, kernel trick implicitly maps the data into a higher dimensionality space where linear separation is possible. This enables the model to create more flexible decision boundaries without explicitly calculating the higher dimensional representation.

The RBF kernel is one of the most commonly used kernels. By using the distance between data points, The RBF kernel allows the model to establish non-linear and curved decision boundaries This enables the SVM to fit more complex data distributions and discern patterns that a linear model is unable to capture.

Figure 3: RBF kernel producing a non-linear decision boundary.

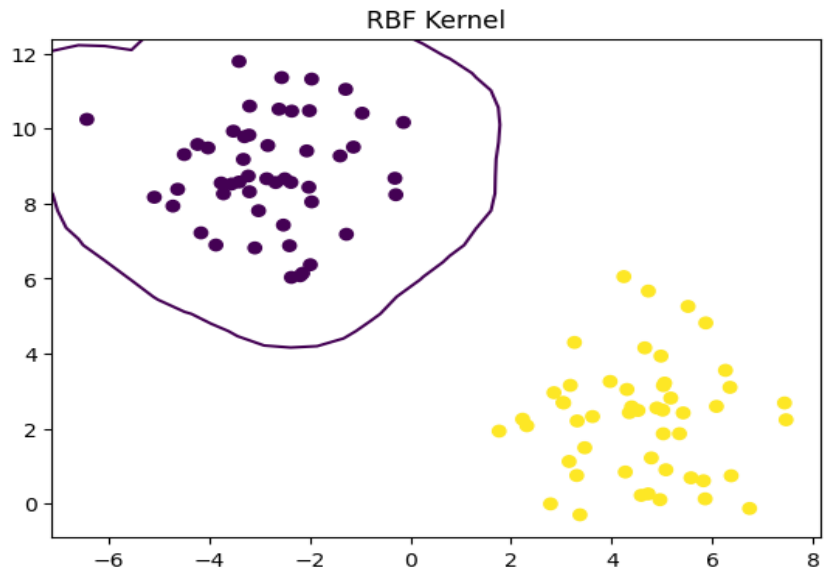
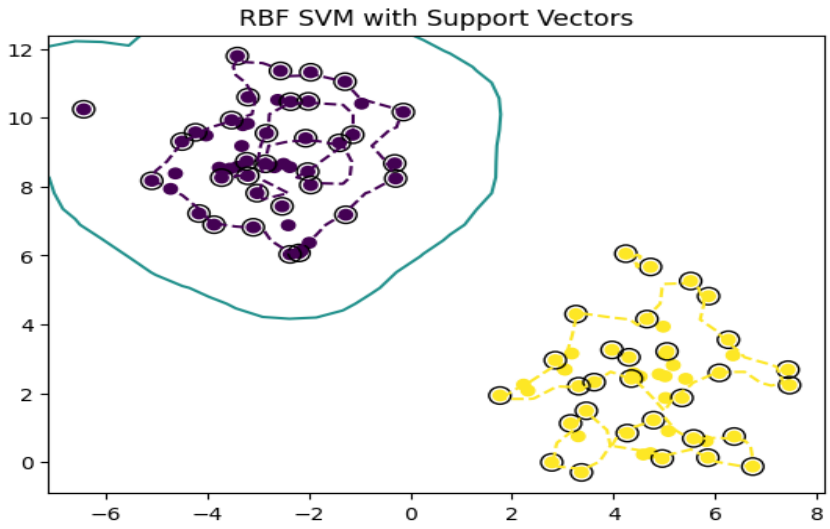


Figure 4: RBF SVM with support vectors illustrating a flexible decision boundary.



Kernel functions perform the similarity quantification of data points, which is a fundamental principle in Support Vector Machines. For the case of the RBF kernel, this notion of similarity is based on how far apart points are in feature space. Points that are near each other are viewed as more related,

while those further apart are treated as less so. This enables the model to construct localised decision regions instead of using a single global linear boundary.

RBF Kernel: Each point of data, called the centre, radiates a bell-like structure that makes influence on other points around it and helps to separate different classes on our dataset gamma another hyperparameter controlling distance. A small value of γ creates a smoother decision boundary since each point influences a larger area; On the other hand, very large value of gamma creates a very complex boundary which might overfit to training data. Hence, the choice of gamma value is key to balancing model flexibility and generalisation.

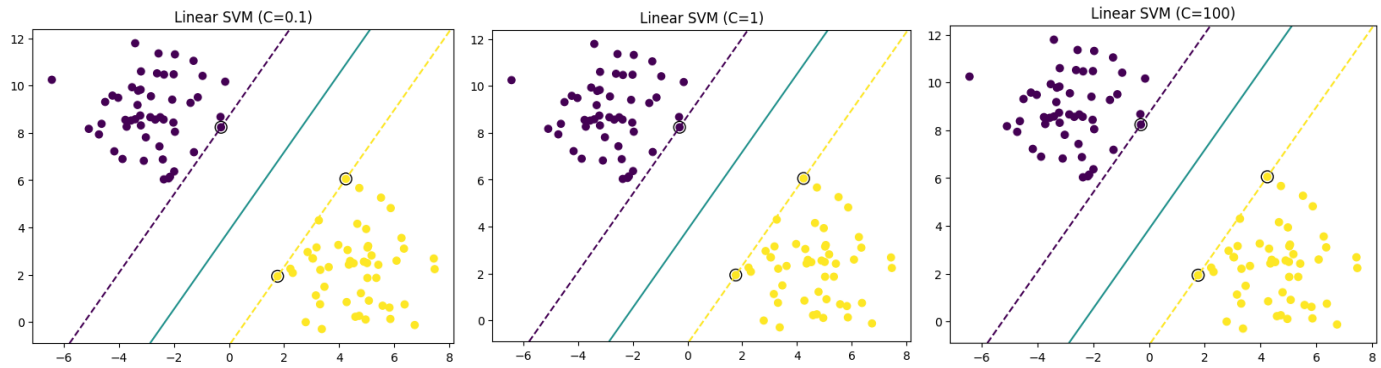
Kernel methods allow for a computation-efficient generalized transformation of data to higher-dimensional spaces without the need to explicitly calculate the mapping, which forms the basis of SVM's efficiency. Especially when you have large datasets or high-dimensional feature spaces. This makes kernel-based SVMs a flexible and robust approach to numerous classification challenges while preserving solid theoretical principles.

The RBF kernel forms a curved decision boundary, as can be seen in Figure 2. In contrast to the linear SVM, which can only find a straight boundary between classes, an RBF-based model is able to warp its decision boundary in order to better separate the two classes. This is why kernel-based SVMs are particularly potent techniques for solving non-linear classification.

Effect of Regularisation Parameter (C)

Besides the kernel choice, the performance of a Support Vector Machine is also greatly affected by the regularisation parameter: C. This value determines how much we are willing to maximise our margin at the cost of misclassifying our training data.

Figure 5,6,7: Effect of different values of the regularisation parameter C on the SVM decision boundary.



A value of c allows a larger margin even if it misclassifies some data points. This results in a simpler decision boundary which generalises better to unseen data but can Underfit on training data. A small value of C allows for a wider margin, creating a less complex decision boundary as it tolerates misclassifications, while a large value of C heavily penalizes misclassification and creates a more

complex decision boundary. Though this can enhance training accuracy, it more than doubles the likelihood of overfitting.

Figure 3 shows how different values of C affect the model. Model allows more margin and some misclassification, so it produces a softer boundary when C = 0.1 C = 1: Creates a balance between the margin size and classification accuracy. But with C = 100, the model tries to classify all points correctly and thus squeezes the boundary up against the data.

This behaviour emphasises the need for choosing a good value of C as it directly influences how well the model generalizes. An appropriate value strikes a balance between bias and variance, allowing the SVM to perform well on both training data and unseen new data.

Case Study: Non-linear Data

Understanding the effectiveness of the RBF kernel in this case can be done by considering how the data gets mapped into a higher-dimensional space. This allow the SVM to find one hyperplane in the original data that separates these two classes. implies the kernel trick, allowing a model to function as though its data were mapped to this new space without directly computing it.

This point illustrates an important feature of SVMs: they can produce very complex decision boundaries without having to resort to extensive manual manipulation of the input features. It can be difficult and time-consuming to design features that capture non-linear relationships in many machine learning tasks. This is where Kernel methods shine, as they take in input data and perform mathematical transformations that automatically capture these relationships.

“On the other hand, that flexibility creates challenges, too. A class of models Part 2: Kernel based Models, C and gamma are very sensitive parameters. Moreover, interpretability may also be harder as the model becomes more complex than the linear models.

Nonetheless, the findings from this case study clearly show that kernel-based support vector machines are very powerful classifiers if there exist substantial non-linearly separable data. They are a powerful technique for handling both structured and unstructured data.

More broadly, many data in real-world applications are not linearly separable (namely, there is no straight-line decision boundary that can effectively separate the classes). As an example demonstrating this limitation, consider a synthetic dataset of concentric circles in which one class encompasses another. This kind of structure creates a challenge for linear classifiers like the standard SVM.

The data set is shown in circular distribution as shown in figure 5, one class forms an inner cluster and the other class surrounds it. The arrangement [[0, 1], [2, 3]] makes it impossible to separate the two classes with a single line without major misclassification.

So when we use a linear SVM on this data-set, our decision boundary does not represent the underlying structure of the data. Figure 6 illustrates where the model tries to segment the classes using a straight line, resulting in misclassification. This exemplifies the inherent limitation of linear SVMs with complex, non-linear data.

We apply the RBF kernel to solve this problem. The RBF kernel allows the model to have a non-linear decision boundary takes the shape of the data by implicitly projecting the input into a higher-dimensional space. As demonstrated in Figure 7, the boundary produced is circularly separable and correctly separates both classes.

Figure 8: Non-linearly separable dataset with circular structure.

Non-linear dataset (circles)

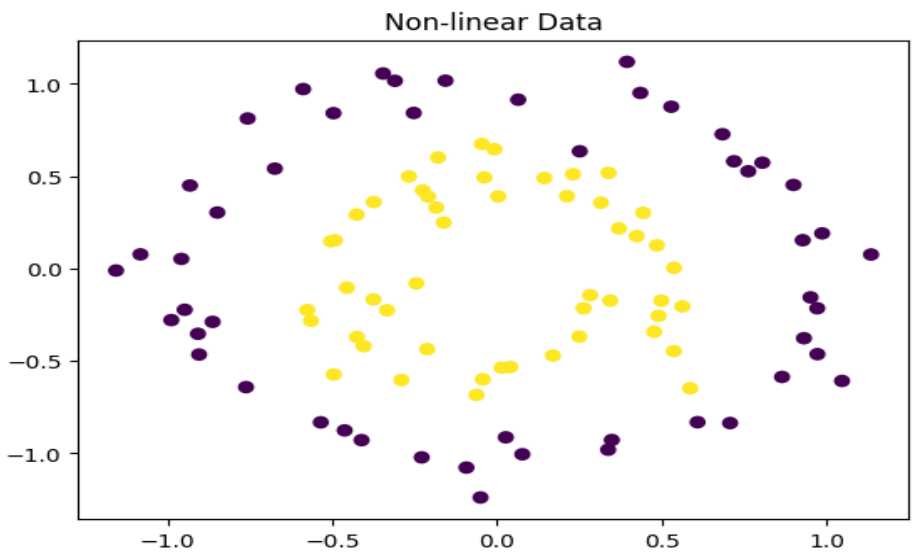


Figure 9: Linear SVM failing to correctly classify non-linear data.

Linear Kernel (Fails)

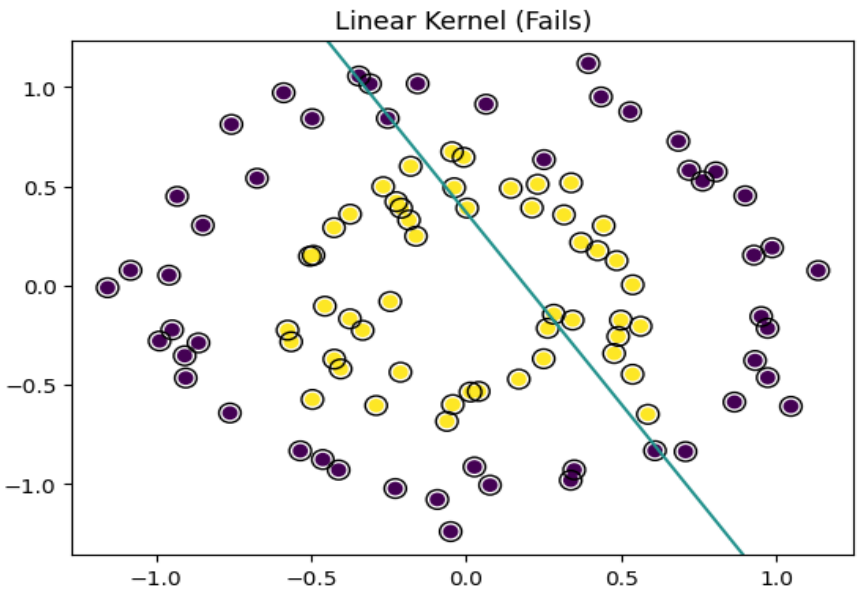
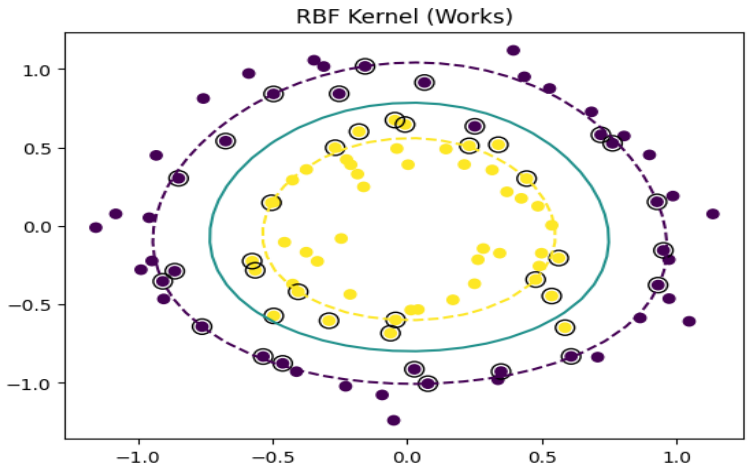


Figure 10: RBF SVM successfully separating non-linear data.

RBF Kernel (Works)



This shows how strong kernel based methods are inside SVM. SVMs can be adapted to many different patterns by using kernels, despite the limited power of a linear model for handling complex data distributions. This renders them especially capable when dealing with real world problems where feature relationships are rarely linear.

Conclusion

We have covered the basic principles and behaviour of Support Vector Machines through a series of experiments and visualisations in this report. Starting with the linear SVM, we showcased how the model finds an optimal decision boundary by maximising the margin between classes, and that this boundary is determined by a few select instances: support vectors. This worked well for simple, linearly separable data at first but soon ran into the limitations of this approach when facing more complex datasets.

To overcome these limitations, kernel methods were introduced where we focus on the Radial Basis Function (RBF) Kernel. Abstract: The Support Vector Machine (SVM) is one of the most powerful learning models available today, capable of representing any decision boundary that separates data belonging to different classes. This greatly improves the model's ability to learn multiple complex patterns.

The effect of the regularisation parameter C was also explored as a balance between maximising margin and minimising classification errors. This parameter tuning controls the generalising capability of model, so it emphasizes the importance of paramater selection.

Lastly, the case study on non-linear data clearly demonstrated that linear SVMs fail to capture complex structures, but kernel-based SVMs could adapt and provide accurate classification. SVMs are among the most powerful and versatile machine-learning techniques, especially when combined with suitable kernel functions and hyperparameter tuning.

References

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